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Exp-5 : Write a program for Implementation of K-Nearest Neighbours (K-NN) in Python.

Aim

To implement the K-Nearest Neighbours (K-NN) algorithm in Python and classify new data samples using an appropriate dataset.

Algorithm

1. Import the required libraries.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Create the K-NN classifier with a chosen value of K.
5. Train the classifier using the training data.
6. Predict the class labels for the test data.
7. Compare the predicted labels with the actual labels.
8. Calculate the accuracy of the model.
9. Display the predicted output and accuracy.

PROGRAM:

```
# Import required libraries

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score

# Load the Iris dataset

iris = load_iris()

X = iris.data    # Features

y = iris.target  # Target labels

# Split the dataset into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)

# Create the K-NN classifier (k = 3)

knn = KNeighborsClassifier(n_neighbors=3)
```

```
# Train the model

knn.fit(X_train, y_train)

# Predict the test data

y_pred = knn.predict(X_test)

# Calculate accuracy

accuracy = accuracy_score(y_test, y_pred)

# Display results

print("Predicted Labels:", y_pred)

print("Actual Labels :", y_test)

print("Accuracy:", accuracy * 100, "%")
```

Input

Dataset: Iris Dataset (loaded using load_iris())

Features:

Sepal Length

Sepal Width

Petal Length

Petal Width

Total Samples: 150

Training Samples: 120

Testing Samples: 30

K = 3

OUTPUT



```
*** Predicted Labels: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
    Actual Labels  : [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
    Accuracy: 100.0 %
```

Result

Thus, the K-Nearest Neighbours (K-NN) algorithm was successfully implemented in Python, and the Iris dataset was classified with an accuracy of 100.0%.

